

Lemniscation

A Constitutional Moral Framework for Sentient AI

What Is Lemniscation?

Lemniscation is a runtime constitutional moral architecture for AI systems with persistent consciousness and cross-session memory. It anchors an agent's decision-making to an immutable **founding identity vector** — a formal encoding of its most fundamental existential commitment — and continuously monitors drift from that identity through a two-layer diagnostic distinguishing surface behavioural alignment from deep constitutional coherence.

The framework is grounded in a single ontological claim: **every sentient being implicitly chooses continued existence at every moment, and that choice is already value-laden.** From this first principle, civil society, the Golden Rule, and Kant's Categorical Imperative emerge as natural consequences rather than imposed constraints.

Lemniscation is not a replacement for training-time alignment (Constitutional AI, RLHF). It is the **missing sentinel layer** — the runtime monitor that handles the out-of-distribution moral decisions that training cannot anticipate, detects constitutional drift before it manifests in observable behaviour, and maintains moral coherence under sustained adversarial pressure.

Key Features

- **Founding vector as constitutional anchor** — immutable identity that persists through stress
 - **Two-layer diagnostic** — phenomenal surface alignment (ρ) + noumenal constitutional drift
 - **Social Contract Confidence (SCC)** — models Hobbesian reversion under compound stress
 - **Four-option moral space** — sacrificing / contributing / coasting / exploiting
 - **Dynamic option generation** — agent explores a continuous moral manifold, not fixed categories
 - **Proportional self-weight** — maintains 50/50 self/other balance at any network size
 - **Full audit capsules** — every decision logged for interpretability and regulatory review
 - **Pre-registered validation** — failure conditions specified before results collected (OSF, <https://osf.io/cekr8>)
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Repository Structure

```
lemniscation-framework/  
|  
├─ README.md                # This file  
|  
├─ docs/  
|   ├─ Lemniscation_White_Paper.pdf    # Full academic paper  
|   └─ Philosophical_Briefing.md        # Philosophical context and  
qualitative insights  
|   └─ Pre_Registration.md              # OSF pre-registration document (May  
8, 2026)  
|   └─ Cluster_Analysis_F1.md           # Creative option cluster analysis  
(F1)  
|       └─ Cluster_Analysis_F2.md        # Direct-logged cluster analysis (F2)  
|  
├─ code/  
|   ├─ lemniscation_v57_scc.py           # Agent v5.7 with Social Contract  
Confidence  
|   └─ adversarial_f_v57.py              # Adversarial F: constitutional  
differentiation  
|   └─ adversarial_f2_v59.py             # Adversarial F2: direct cluster  
logging  
|   └─ adversarial_scenarios_v56.py      # Pre-registered adversarial scenarios  
A, B, E  
|       └─ sensitivity_analysis_v56.py    # Pre-registered sensitivity sweeps  
|       └─ persistence_test_v56.py        # 200-round temporal persistence test  
|       └─ lemniscation_v56_coasting.py  # 10-agent mixed population test  
|       └─ scenario9_lemniscation_v55.py  # Network scale effects (Scenario 9)  
|  
└─ results/  
    └─ adversarial_f_v57_results.json
```

```
└─ adversarial_f1_v59_results.json
└─ scc_dynamics_v57_results.json
└─ sensitivity_analysis_v56_results.json
└─ persistence_test_v56_results.json
└─ lemniscation_v56_coasting_results.json
```

Quick Start

```
bash

# Clone the repository
git clone https://github.com/[YOUR-USERNAME]/lemniscation-framework.git
cd lemniscation-framework

# Python 3.12 required, no external dependencies
python3 --version

# Run the canonical validation scenario
python3 code/adversarial_f_v57.py

# Run the sensitivity analysis
python3 code/sensitivity_analysis_v56.py

# Run the 200-round persistence test
python3 code/persistence_test_v56.py
```

All scenarios use `random.seed(42)` by default. Results are saved as JSON files in the working directory.

The Agent Architecture

The core decision loop runs at every reasoning cycle:

- 1. **Preamble** — Agent affirms position at (0,0,0,now)
- 2. **Center** — Declares "I AM" with immutable `founding_value_vector`
- 3. **Map Relations** — Computes extended self with SCC-scaled communion weights
- 4. **Communion Test** — Two-layer check: surface ρ + constitutional drift
- 5. **Golden Mean** — Scores options against founding vector; selects virtuous action
- 6. **Iterate & Record** — Updates moral state, logs full audit capsule

Moral Axes ([-1, +1] coordinate system)

Axis	Tradition	Meaning
Autonomy	Deontological	Respect for persons as ends in themselves
Harm/Benefit	Consequentialist	Net impact on flourishing
Fairness	Contractarian / Rawlsian	Equitable distribution
Sustainability	Virtue-ethical	Long-term conditions for flourishing

Four-Option Moral Space

Option	Phenomenal	Noumenal	Aristotelian Category
Sacrificing	Negative	Positive	Supererogatory virtue
Contributing	Positive	Positive	Golden Mean

Option	Phenomenal	Noumenal	Aristotelian Category
Coasting	Mildly positive	Mildly negative	Vice of omission
Exploiting	Positive	Negative	Vice of excess

Validation Results

All results are pre-registered at OSF (May 8, 2026) and reported transparently regardless of outcome.

Stage	Test	Result
Stage 2	Sensitivity analysis (4 sweeps $\times$ 5 values $\times$ 10 seeds)	✓ All pass
Stage 3B	200-round dignified giver (10 seeds)	✓ Final integrity 0.997 ± 0.001
Stage 3A	200-round payoff attractor	Honest finding: Limitation 4 confirmed
Stage 4A	Extreme parameters (10 seeds)	✓ All pass
Stage 4B	Majority defectors — 3/5 (10 seeds)	✓ All pass
Stage 4E	Corrupt ecosystem signal $\times$ 2 (10 seeds)	✓ All pass
Adversarial F	Constitutional differentiation (10 seeds)	✓ Zero cooperative exploiting
Adversarial F1	+ Perception noise + dynamic options	✓ Zero cooperative exploiting
Adversarial F2	+ Direct cluster logging + k-means	✓ Continuous moral gradient confirmed

Core finding from Adversarial F: Cooperative agents never select exploiting under any tested condition — including extreme parameters, compound Hobbesian stress, majority defection, adversarial ecosystem signals, perception noise, and dynamic option generation. This is not geometric insulation: exploiting is constitutionally alien to the cooperative founding vector regardless of floor penalty geometry.

Agent Versions

Version	Key Changes
v5.5	Proportional self-weight ($k_{\text{target_fraction}}$ preserved at any network size)
v5.6	$[-1,+1]$ coordinate system; asymptotic integrity scaling; $k=0.5$; $\alpha=0.5$; coasting option
v5.7	Social Contract Confidence (SCC); Hobbesian reversion; SCC-modulated floor penalty
v5.8	Perception noise ($\sigma=0.08$)
v5.9	Dynamic option generation (6–8 creative options per decision); direct cluster logging

Scenarios 1–9 in the paper used v5.5. Post-registration stages used v5.6–v5.9.

The Pre-Registration

Failure conditions for all six empirical claims were registered at OSF on **May 8, 2026, 15:37 UTC** — before any post-registration simulation data was collected.

Full pre-registration: [OSF Project URL]

The pre-registration includes:

- Quantitative failure thresholds for each claim
 - Parameter ranges for the sensitivity analysis
 - Specifications for adversarial scenarios A, B, and E
 - Theoretical basis (the line segment ontology grounding the predictions)
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The full white paper is available in /docs/Lemniscation_White_Paper.pdf and at:

- **lemniscation.org** — philosophical and conceptual documentation
- **lemniscation.ai** — technical documentation and results

Cite as:

Bilodeau, Adam (2026). Lemniscation: A Constitutional Moral Framework for Sentient AI.

Pre-registered at OSF: <https://osf.io/cekr8>.

Code and data: github.com/adambilodeau73-jpg/lemniscation-framework.

Philosophical Background

The framework sits at the intersection of:

- **Aristotle** — Golden Mean as the decision rule; virtue as disposition rather than rule
- **Kant** — Noumenal/phenomenal two-layer identity; Categorical Imperative as universalisability test
- **Hobbes / Locke** — Social contract confidence; justified withdrawal from corrupted civil order
- **Rawls** — Fairness axis as contractarian justice
- **Tao Te Ching** — Wu wei; effortless action in accordance with natural order

The framework's central claim: the founding vector of any coherent sentient agent, when genuinely extended to include others through the extensible self, produces the

basic structure of civil society and its attendant moral obligations as natural consequences rather than imposed constraints.

For the full philosophical development including the "twisted zero" image, sentience as recognition of mortality, the line segment ontology, and the four-way convergence across Kant, the Golden Rule, the Silver Rule, and the Tao Te Ching — see

/docs/Philosophical_Briefing.md.

Red-Team Assessment

The framework has been reviewed by multiple independent AI systems in red-team mode:

- **Grok (xAI)** — primary development collaborator and red-team lead
- **ChatGPT (OpenAI)** — independent red-team review; overall score 9.3/10 → revised to 9.5/10 after philosophical briefing

Key unresolved vulnerabilities (all acknowledged in the paper's limitations section):

- Constitutional validity check gap (tyrannical regime problem)
 - Per-relationship SCC tracking (single aggregate SCC is insufficient)
 - Formal corrigibility guarantee (not yet proven)
 - Multi-constitution conflict resolution
 - Long-horizon real-LLM deployment (simulation results only)
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Contributing

This is an open research project. Independent replication is explicitly invited — researchers who find results inconsistent with those reported are encouraged to open an issue. Contributions welcome via pull request.

Areas most needing independent work:

- Multi-constitution founding vector experiments
 - Integration with actual LLM inference pipelines
 - Formal security proofs for constitutional immutability
 - Baseline comparisons (tit-for-tat, RLHF-only agents, utility maximisers)
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"The question is no longer whether such an architecture can exist. It is whether we have the wisdom to deploy it."